

Exploratory Data Analysis on Titanic Passengers

Introduction:

The dataset for exploratory data analysis was done for the Titanic dataset. A specific individual did not formulate it but took from records of the history of passengers that were onboard the ship, the Titanic.

The access to this dataset was taken from the popular machine learning competition website, Kaggle and it can also be accessed from other platforms like online repositories and educational websites. The dataset could answer questions about demographics, rate of survival, relationship, fare and class, gender and survival, age and survival, embarkation points, information on the cabin, analysis of name, and analysis of socioeconomics. The basic fitness of the dataset was assessed. The total size of the dataset was 15708 and four columns had their values missing.

Data Pre-processing and Statistical Interpretation:

Data Cleaning:

The data are missing on the dataset.

This is due to vivid reasons. Some of them being due to data entry errors whereby clerk might have forgotten to input a value, incomplete data collection, technological issues, non-applicable information(cabin variable might be missing for the passengers who didn't get cabin), privacy and confidentiality, time-related factors and data cleaning and processing etc.

Median method is used for data imputation in this case as it is quick way to fill missing values as the values in age and fare are missing at random (MAR).

Mode method is used for data imputation in this case as this is categorical variable (take on one of a limited and usually fixed number of values).

As the missing values of cabin column is as high as 77.46 % and it has significant part of data missing, filling up these values may cause uncertainty and bias in the analysis.

Size before data cleaning: 15708

Size after data cleaning: 14399

Summary Statistics

Analyzing the summary form describe():

PassengerId: The value goes from 1 to 1309 which indicates 1309 observations.

Survived: Mean is 0.38, thus meaning 38% of passengers survived and standard deviation of 0.48 indicates the variability in survival rates.

Pclass: The mean is approximately 2.29 and the value ranges from 1 to 3 which is to say on average passengers were in the second class.

SibSp: Mean being 0.50, which means less than one sibling or spouse is with the passenger. Maximum number is 8, which means few passengers had large number of siblings or spouse present.

Parch: The mean being 0.39 which means less than one parent or child is with the passenger. The maximum value is 9 which suggests few passengers had a large numbers or parents on the ship.

Fare: The minimum fare is 0, suggesting some passengers may had traveled without buying the tickets. The standard deviation is 51.74 which suggests a significant variability in the prices of fares.

Likewise for categorical column unique values for sex and embarked columns were found out by using .unique() function. Their mode were also found out through .mode().

Data visualization and Explorations:

Univariate Analysis

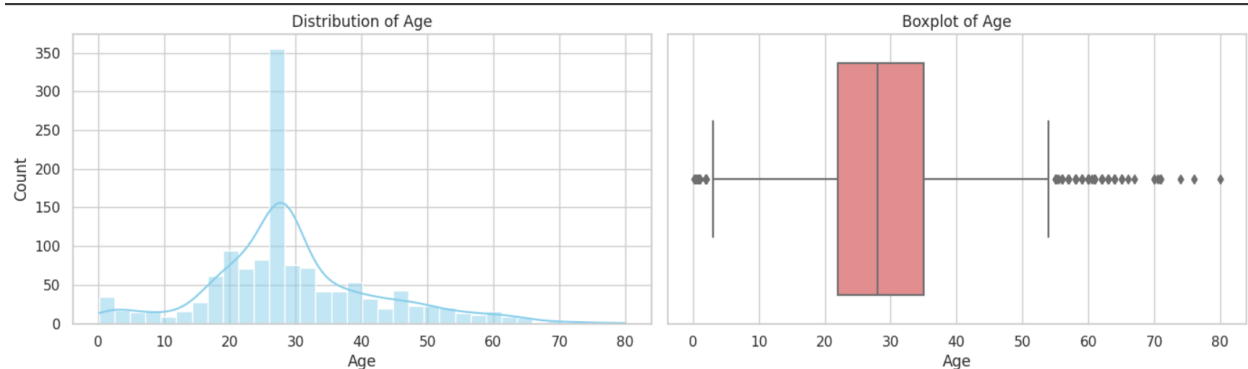


Figure 1 Distribution of Age

Distribution of Age: The line chart depicts the distribution of ages of passengers' ages which shows the density of value of age. Boxplot summarizes central tendency spread and presence of outliers of the age distribution. Age distribution is slightly right-skewed and has peak in young adult age.

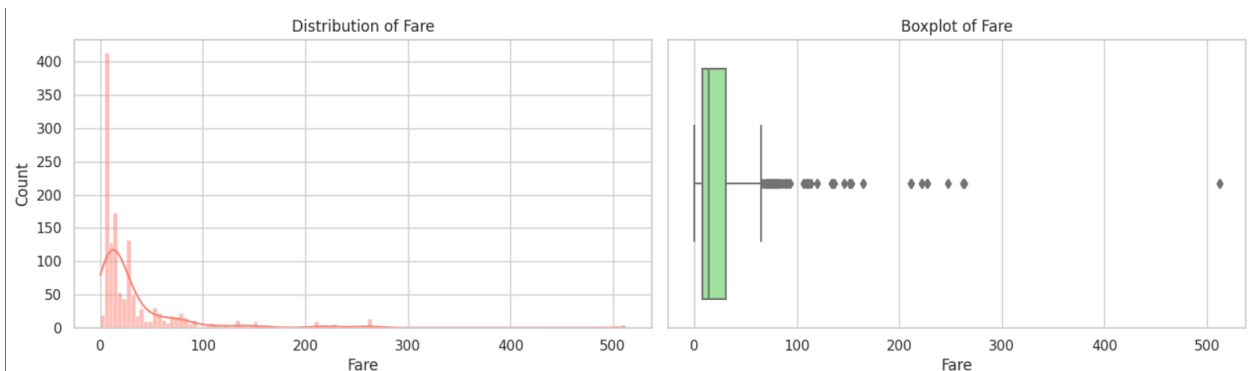


Figure 2 Distribution of Fare

Distribution of Fare: The bar chart shows frequency of fare values. The boxplot summarizes central tendency and potential outliers of fare distribution. The majority of passengers paid lower fares thus right-skewed. Boxplot shows the presence of less high outliers which means few paid relatively larger fares.

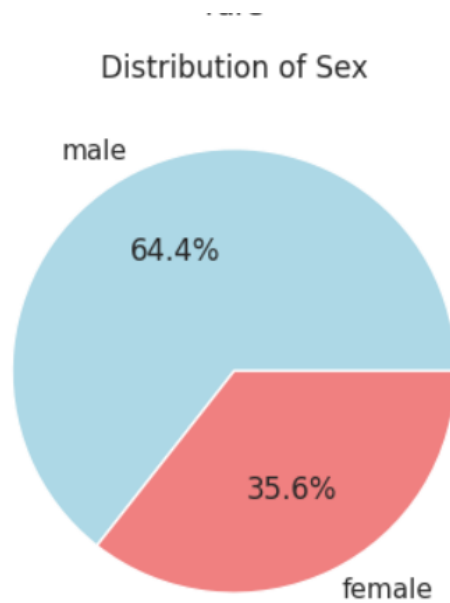


Figure 3 Distribution of Sex

Distribution of Sex: Pie chart shows there were more males compared to females on the Titanic which is shown by larger blue cut.

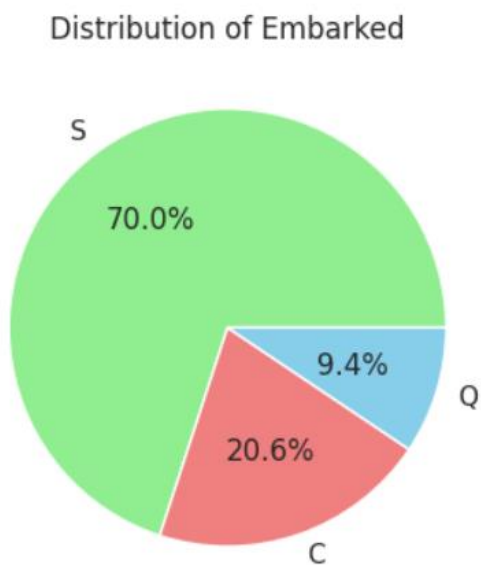


Figure 4 Distribution of Embarked

Distribution of Embarked: The pie chart depicts embarkation ports, largest segment being from Southampton(S) then Cherbourg(C) comes second. Nevertheless, there were fewest passengers from Queenstown (Q).

Bivariate Analysis



Figure 5 Pairplot of Numeric Variables

Pair plot of Numeric Variables: The pair plot shows scatter plots for every pair of variable of numerical in the dataset. Diagonal elements shows the distribution of single variables. It allows to inspect correlations and relations between numeric variables. It shows how the variables are related. In the plot age vs. fare there is not any linear relation between age and fare. Individual characters is shown by diagonal histogram of each variable.



Figure 6 Correlation Heat map of Numeric Variables

Correlation Heat map of Numeric Variables: Represents the correlation matrix of numeric variables. Colors depicts the strength and direction of correlation between variables. Darker colors show stronger positive correlations and lighter color mean negative correlation. Pclass and Fare have strong correlation. This provide a quantitative view of correlation.

The reader can distinguish the variables strongly correlated and find patterns and dependencies. Correlation can guide feature selection.

Conclusion and Reflect:

The structure of the dataset along with its characters can be visualized and known clearly through EDA (Exploratory Data Analysis). Patterns, potential outliers, and relationships can be seen through pair plots and heat maps. Visualization from line plots, bar graphs, and pie charts can save time and make it easier to have a summary from a large amount of data which is helpful for decision making. The interpretation as well as visualizations need consideration of the context and specific aims. Effective handling missing data and outliers in different scenarios still remains my question. Likewise, I am interested on more advanced techniques for processing the data for effective data reading. Being first project of this kind, it was relatively harder as well as a great learning process.